

Trust where payments depended in a complex way on the 30-day commercial paper rate, a 30-year Treasury bond price, and the yield on a 5-year Treasury bond.

SUMMARY

The two most common types of swaps are interest rate swaps and currency swaps. In an interest rate swap, one party agrees to pay the other party interest at a fixed rate on a notional principal for a number of years. In return, it receives interest at a floating rate on the same notional principal for the same period of time. In a currency swap, one party agrees to pay interest on a principal amount in one currency. In return, it receives interest on a principal amount in another currency.

Principal amounts are not exchanged in an interest rate swap. In a currency swap, principal amounts are usually exchanged at both the beginning and the end of the life of the swap. For a party paying interest in the foreign currency, the foreign principal is received, and the domestic principal is paid at the beginning of the life of the swap. At the end of the life of the swap, the foreign principal is paid and the domestic principal is received.

An interest rate swap can be used to transform a floating-rate loan into a fixed-rate loan, or vice versa. It can also be used to transform a floating-rate investment to a fixed-rate investment, or vice versa. A currency swap can be used to transform a loan in one currency into a loan in another currency. It can also be used to transform an investment denominated in one currency into an investment denominated in another currency.

The interest rate and currency swaps considered in main part of the chapter can be regarded portfolios of forward contracts. They can be valued by assuming the forward interest rates and exchange rates observed in the market today will occur in the future.

FURTHER READING

- Alm, J., and F. Lindskog, "Foreign Currency Interest Rate Swaps in Asset-Liability Management for Insurers," *European Actuarial Journal*, 3 (2013): 133–58.
- Corb, H. *Interest Rate Swaps and Other Derivatives*. New York: Columbia University Press, 2012.
- Flavell, R. *Swaps and Other Derivatives*, 2nd edn. Chichester: Wiley, 2010.
- Johannes, M., and S. Sundaresan, "The Impact of Collateralization on Swap Rates," *Journal of Finance*, 61, 1 (February 2007): 383–410.
- Litzenberger, R. H. "Swaps: Plain and Fanciful," *Journal of Finance*, 47, 3 (1992): 831–50.
- Mommel, C., and A. Schertler. "Bank Management of the Net Interest Margin: New Measures," *Financial Markets and Portfolio Management*, 27, 3 (2013): 275–97.
- Purnananandan, A. "Interest Rate Derivatives at Commercial Banks: An Empirical Investigation," *Journal of Monetary Economics*, 54 (2007): 1769–1808.

Short Concept Questions

- 7.1. Explain how payments are calculated in an OIS.
- 7.2. Explain how the risk-free zero curve is calculated from OIS rates.

- 7.3. How many exchanges are usually made in a one-year OIS swap?
- 7.4. Explain whether principal is exchanged in (a) an interest rate swap and (b) a currency swap.
- 7.5. A company has a five-year investment earning 3% per year. Five-year interest rate swaps are quoted as bid 3.21%, ask 3.25%. What floating investment rate investment can be achieved with a swap?
- 7.6. A company can borrow for five years at 4%. Five-year interest rate swaps are quoted as bid 3.21%, ask 3.25%. What floating borrowing rate can be obtained using a swap?
- 7.7. Explain why borrowing at a floating rate and swapping to fixed does not necessarily give rise to borrowing at the same fixed rate every year.
- 7.8. What estimates have to be made to value an interest rate swap?
- 7.9. Consider a swap when fixed is paid and floating is received. If the term structure of interest rates is upward sloping, do the early exchanges have a positive or negative value?
- 7.10. Explain two ways a fixed-for-fixed currency swap can be valued.

Practice Questions

- 7.11. Companies A and B have been offered the following rates per annum on a \$20 million five-year loan:

	<i>Fixed rate</i>	<i>Floating rate</i>
Company A	5.0%	SOFR + 0.1%
Company B	6.4%	SOFR + 0.6%

Company A requires a floating-rate loan; Company B requires a fixed-rate loan. Design a swap that will net a bank, acting as intermediary, 0.1% per annum and that will appear equally attractive to both companies.

- 7.12. A \$100 million interest rate swap has a remaining life of 10 months. Under the terms of the swap, six-month LIBOR is exchanged for 4% per annum (compounded semiannually). Six-month LIBOR forward rates for all maturities are 3% (with semiannual compounding). The six-month LIBOR rate was 2.4% two months ago. OIS rates for all maturities are 2.7% with continuous compounding. What is the current value of the swap to the party paying floating? What is the value to the party paying fixed?
- 7.13. Company X wishes to borrow U.S. dollars at a fixed rate of interest. Company Y wishes to borrow Japanese yen at a fixed rate of interest. The amounts required by the two companies are roughly the same at the current exchange rate. The companies have been quoted the following interest rates, which have been adjusted for the impact of taxes:

	<i>Yen</i>	<i>Dollars</i>
Company X	5.0%	9.6%
Company Y	6.5%	10.0%

Design a swap that will net a bank, acting as intermediary, 50 basis points per annum. Make the swap equally attractive to the two companies and ensure that all foreign exchange risk is assumed by the bank.

- 7.14. A currency swap has a remaining life of 15 months. It involves exchanging interest at 10% on £20 million for interest at 6% on \$30 million once a year. The term structure of risk-free interest rates in the United Kingdom is flat at 7% and the term structure of risk-free interest rates in the United States is flat at 4% (both with annual compounding). The current exchange rate (dollars per pound sterling) is 1.5500. What is the value of the swap to the party paying sterling? What is the value of the swap to the party paying dollars?
- 7.15. Explain the difference between the credit risk and the market risk in a swap.
- 7.16. A corporate treasurer tells you that he has just negotiated a five-year loan at a competitive fixed rate of interest of 5.2%. The treasurer explains that he achieved the 5.2% rate by borrowing at a six-month floating reference rate plus 150 basis points and swapping the floating reference rate for 3.7%. He goes on to say that this was possible because his company has a comparative advantage in the floating-rate market. What has the treasurer overlooked?
- 7.17. A bank enters into an interest rate swap with a nonfinancial counterparty using bilaterally clearing where it is paying a fixed rate of 3% and receiving floating. No collateral is posted and no other transactions are outstanding between the bank and the counterparty. What credit risk is the bank subject to? Discuss whether the credit risk is greater when the yield curve is upward sloping or when it is downward sloping.
- 7.18. Companies X and Y have been offered the following rates per annum on a \$5 million 10-year investment:

	<i>Fixed rate</i>	<i>Floating rate</i>
Company X	8.0%	LIBOR
Company Y	8.8%	LIBOR

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.2% per annum and will appear equally attractive to X and Y.

- 7.19. A financial institution has entered into an interest rate swap with company X. Under the terms of the swap, it receives 4% per annum and pays six-month LIBOR on a principal of \$10 million for five years. Payments are made every six months. Suppose that company X defaults on the sixth payment date (end of year 3) when six-month forward LIBOR rates for all maturities are 2% per annum. What is the loss to the financial institution? Assume that six-month LIBOR was 3% per annum halfway through year 3 and that at the time of the default all OIS rates are 1.8% per annum. OIS rates are expressed with continuous compounding; other rates are expressed with semiannual compounding.
- 7.20. A financial institution has entered into a 10-year currency swap with company Y. Under the terms of the swap, the financial institution receives interest at 3% per annum in Swiss francs and pays interest at 8% per annum in U.S. dollars. Interest payments are exchanged once a year. The principal amounts are 7 million dollars and 10 million francs. Suppose that company Y declares bankruptcy at the end of year 6, when the exchange rate is \$0.80 per franc. What is the cost to the financial institution? Assume that, at the end of year 6, risk-free interest rates are 3% per annum in Swiss francs and 8% per annum in U.S. dollars for all maturities. All interest rates are quoted with annual compounding.

- 7.21. Companies A and B face the following interest rates (adjusted for the differential impact of taxes):

	<i>Company A</i>	<i>Company B</i>
U.S. dollars	Floating + 0.5%	Floating + 1.0%
Canadian dollars	5.0%	6.5%

Assume that A wants to borrow U.S. dollars at a floating rate of interest and B wants to borrow Canadian dollars at a fixed rate of interest. A financial institution is planning to arrange a swap and requires a 50-basis-point spread. If the swap is equally attractive to A and B, what rates of interest will A and B end up paying?

- 7.22. After it hedges its foreign exchange risk using forward contracts, is the financial institution's average spread in Figure 7.11 likely to be greater than or less than 20 basis points? Explain your answer.
- 7.23. "Nonfinancial companies with high credit risks are the ones that cannot access fixed-rate markets directly. They are the companies that are most likely to be paying fixed and receiving floating in an interest rate swap." Assume that this statement is true. Do you think it increases or decreases the risk of a financial institution's swap portfolio? Assume that companies are most likely to default when interest rates are high.
- 7.24. Why is the expected loss to a bank from a default on a swap with a counterparty less than the expected loss from the default on a loan to the counterparty when the loan and swap have the same principal? Assume that there are no other derivatives transactions between the bank and the counterparty, that the swap is cleared bilaterally, and that no collateral is provided by the counterparty in the case of either the swap or the loan.
- 7.25. A bank finds that its assets are not matched with its liabilities. It is taking floating-rate deposits and making fixed-rate loans. How can swaps be used to offset the risk?
- 7.26. Explain how you would value a swap that is the exchange of a floating rate in one currency for a fixed rate in another currency.
- 7.27. OIS rates are 3.4% for all maturities. What is the value of an OIS swap with two years to maturity where 3% is received and the floating reference rate is paid. Assume annual compounding, annual payments, and \$100 million principal.
- 7.28. A financial institution has entered into a swap where it agreed to make quarterly payments at a rate of 3% per annum and receive the SOFR three-month reference rate on a notional principal of \$100 million. The swap now has a remaining life of 7.5 months. Assume the risk-free rates with continuous compounding (calculated from SOFR) for 1.5, 4.5, and 7.5 months are 2.8%, 3.0%, and 3.1%, respectively. Assume also that the continuously compounded risk-free rate observed for the last 1.5 months is 2.7%. Estimate the value of the swap.
- 7.29. (a) Company A has been offered the swap quotes in Table 7.4. It can borrow for three years at 3.45%. What floating rate can it swap this fixed rate into? (b) Company B has been offered the swap quotes in Table 7.4. It can borrow for five years at floating plus 75 basis points. What fixed rate can it swap this rate into? (c) Explain the rollover risks that Company B is taking.
- 7.30. The one-year LIBOR rates is 3%, and the LIBOR forward rate for the 1- to 2-year period is 3.2%. The three-year swap rate for a swap with annual payments is 3.2%. What is the LIBOR forward rate for the 2- to 3-year period if OIS zero rates for maturities of one,

two, and three years are 2.5%, 2.7%, and 2.9%, respectively. What is the value of a three-year swap where 4% is received and LIBOR is paid on a principal of \$100 million. All rates are annually compounded.

- 7.31. A financial institution has entered into a swap where it agreed to receive quarterly payments at a rate of 2% per annum and pay the SOFR three-month reference rate on a notional principal of \$100 million. The swap now has a remaining life of 10 months. Assume the risk-free rates with continuous compounding (calculated from SOFR) for 1 month, 4 months, 7 months, and 10 months are 1.4%, 1.6%, 1.7%, and 1.8%, respectively. Assume also that the continuously compounded risk-free rate observed for the last two months is 1.1%. Estimate the value of the swap.
- 7.32. Company A, a British manufacturer, wishes to borrow U.S. dollars at a fixed rate of interest. Company B, a U.S. multinational, wishes to borrow sterling at a fixed rate of interest. They have been quoted the following rates per annum:

	<i>Sterling</i>	<i>U.S. Dollars</i>
Company A	11.0%	7.0%
Company B	10.6%	6.2%

(Rates have been adjusted for differential tax effects.) Design a swap that will net a bank, acting as intermediary, 10 basis points per annum and that will produce a gain of 15 basis points per annum for each of the two companies.

- 7.33. The five-year swap rate when cash flows are exchanged semiannually is 4%. A company wants a swap where it receives payments at 4.2% per annum on a notional principal of \$10 million. The OIS zero curve is flat at 3.6%. How much should a derivatives dealer charge the company. Assume that all rates are expressed with semiannual compounding and ignore bid–ask spreads.